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PART A:One-Way ANOVA Analysis

Descriptive Statistics (Table 1)

```python
import pandas as pd
import numpy as np

df = pd.read_excel('Study_Habits_data.xlsx')

table1 = df.groupby('Primary_Study_Method')['Final_Course_Percentage'].agg([
    ('n', 'count'),
    ('Mean', 'mean'),
    ('SD', 'std')
]).round(1)
print(table1)
```

```
Microsoft Windows [版本 10.0.26200.7309]
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C:\Users\canana\Desktop\讲义\AAAAA\数据分析（ASM315）\Lab\Lab4>C:/Users/canana/AppData/Local/Programs/Python/Python312/python.exe c:/Users/canana/Desktop/讲义/AAAAA/数据分析（ASM315）/Lab/Lab4/Count.py
                      n  Mean   SD
Primary_Study_Method
Cramming             60  57.2  3.5
Practice_Testing     60  87.9  2.6
Spaced_Repetition    60  79.1  2.6
```

Six-Step Hypothesis Testing Procedure

Step 1: State Hypotheses

H_0: $\mu_1 = \mu_2 = \mu_3$ (No difference in mean final percentage across study methods)

H_1: At least one mean differs

Step 2: Choose Test Statistic

One-way ANOVA F-statistic

Step 3: Select Significance Level

$\alpha = 0.05$

Step 4: Compute Test Statistic

Using Pytho, the ANOVA table is obtained:

```python
from scipy import stats
import statsmodels.api as sm
from statsmodels.stats.multicomp import pairwise_tukeyhsd
import pandas as pd

df = pd.read_excel('Study_Habits_data.xlsx')

# 提取数据
cram = df[df['Primary_Study_Method']=='Cramming']['Final_Course_Percentage']
space = df[df['Primary_Study_Method']=='Spaced_Repetition']['Final_Course_Perc
pract = df[df['Primary_Study_Method']=='Practice_Testing']['Final_Course_Perce

# 方差齐性检验（Levene检验）
levene_stat, levene_p = stats.levene(cram, space, pract)
print(f"Levene检验: p={levene_p:.4f}")

# 单因素ANOVA
f_stat, p_value = stats.f_oneway(cram, space, pract)
print(f"ANOVA结果: F={f_stat:.2f}, p={p_value:.4f}")
```

```
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C:\Users\canana\Desktop\讲义\AAAAA\数据分析（ASM315）\Lab\Lab4>C:/Users/canana/AppData/Local/Programs/Python/Python312/python.exe c:/Users/canana/Desktop/讲义/AAAAA/数据分析（ASM315）/Lab/test_static.py
Levene检验: p=0.0716
ANOVA结果: F=1749.71, p=0.0000
```

Step 5: Decision Rule

Reject H_0 if $p < 0.05$

Step 6: Conclusion

Since $p < 0.001 < 0.05$, we reject the null hypothesis. There is statistically significant evidence that at least one study method leads to a different mean final course percentage.

Part B: Chi-Square Test of Independence

Step 1: State Hypotheses

H_0: Study method and grade category are independent

H_1: Study method and grade category are associated

Step 2: Choose Test Statistic

Chi-square test statistic (χ^2)

Step 3: Select Significance Level

$\alpha = 0.05$

```python
import pandas as pd
df = pd.read_excel('Study_Habits_data.xlsx')


contingency_table = pd.crosstab(
    df['Primary_Study_Method'],
    df['Grade_Category']
)
print("Table 2: 列联表")
print(contingency_table)
```

```
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C:\Users\canana\Desktop\讲义\AAAAA\数据分析（ASM315）\Lagrams/Python/Python312/python.exe c:/Users/canana/Desktop/partB.py
Table 2: 列联表
Grade_Category        Distinction  Fail  Pass
Primary_Study_Method
Cramming                        0    44    16
Practice_Testing               60     0     0
```

Step 4: Compute Test Statistic

$\chi^2 = 180.00$, df = 4, $p < 0.001$

Expected frequencies all >5, meeting test assumption.

Step 5: Decision Rule

Reject H_0 if $p < 0.05$

Step 6: Conclusion

Since $p < 0.001 < 0.05$, we reject the null hypothesis. There is a statistically significant association between study method and final grade category.

Part C: Discussion Questions

1. Compare conclusions from ANOVA and Chi-square analyses

Both tests yielded statistically significant results ($p < 0.001$). The ANOVA showed that study method significantly affects mean numerical scores, while the Chi-square test showed that study method significantly affects the distribution of categorical grade outcomes. The conclusions are consistent: study method has a substantial impact on academic performance, whether measured continuously or categorically.

2. Explain why each test was appropriate for its respective research question

ANOVA was appropriate for Research Question 1 because:

The dependent variable (Final_Course_Percentage) is continuous/interval.We compare means across three independent groupsThe research question asks about differences in average scores

Chi-square test was appropriate for Research Question 2 because:

Both variables (Primary_Study_Method and Grade_Category) are categorical. We examine association/independence between two nominal variables. The research question asks about relationship in frequency distributions.

3. Discuss limitations of the study and potential implications

Limitations:

Sampling: Single university sample limits generalizability

Design: Cross-sectional data cannot establish causality

Variables: Potential confounding factors not controlled

Mesurement: Self-reported study methods may be subject to bias

Simplification: Grade categories collapse continuous performance into broad groups

Potential Implications:

Educational Practice: Results suggest promoting spaced repetition and practice testing over cramming

Student Advising: Academic advisors can recommend evidence-based study strategies

Further Research: Randomized controlled trials could establish causal relationships**Curriculum Design**: Courses could incorporate structured practice testing components